Linked List Practice Problem Set

Curated LeetCode Challenges & Full Linked List Implementation Assignment

Part 1: Comprehensive Implementation Challenge

Assignment Problem: Complete Singly Linked List Implementation

Core Fundamentals

Design and implement a custom `SinglyLinkedList` class in Java/Python/C++ from scratch without using any built-in list data structures. Your implementation must support creation, insertion, deletion, and traversal operations.

Required Operations:

- **Node Creation:** Define a `Node` class storing an integer `val` and reference `next`.
- **Insertion Methods:**
  - `insertAtStart(int val)` — Adds a node at the head (Position 1).
  - `insertAtEnd(int val)` — Appends a node at the tail.
  - `insertAtPos(int val, int pos)` — Inserts a node at 1-based index `pos`.
- **Deletion Methods:**
  - `deleteAtStart()` — Removes the head node.
  - `deleteAtEnd()` — Removes the last node.
  - `deleteAtPos(int pos)` — Removes the node at 1-based index `pos`.
- **Traversal Method:** `display()` — Prints all node values separated by `"->"` ending in `"null"`.

Sample Test Workflow:

```
LinkedList list = new LinkedList(); list.insertAtStart(10); // List: 10 -> null
list.insertAtEnd(30); // List: 10 -> 30 -> null list.insertAtPos(20, 2); // List:
10 -> 20 -> 30 -> null list.display(); // Output: 10 -> 20 -> 30 -> null
list.deleteAtStart(); // List: 20 -> 30 -> null list.deleteAtEnd(); // List: 20 ->
null list.display(); // Output: 20 -> null
```

Part 2: Essential LeetCode Practice Questions

1. LeetCode #206: Reverse Linked List

Easy

Given the `head` of a singly linked list, reverse the list, and return *the reversed list's head*.

Example: Input: `head = [1,2,3,4,5]` | Output: `[5,4,3,2,1]`

🔗 Open Problem on LeetCode (206)

2. LeetCode #876: Middle of the Linked List

Easy

Given the `head` of a singly linked list, return *the middle node of the linked list*. If there are two middle nodes, return the second middle node.

Example: Input: `head = [1,2,3,4,5,6]` | Output: `Node with value 4`

🔗 Open Problem on LeetCode (876)

3. LeetCode #83: Remove Duplicates from Sorted List

Easy

Given the `head` of a sorted linked list, delete all duplicates such that each element appears only once. Return *the linked list sorted as well.*

Example: Input: `head = [1,1,2,3,3]` | Output: `[1,2,3]`

🔗 Open Problem on LeetCode (83)

4. LeetCode #2181: Merge Nodes in Between Zeros

Medium

You are given the head of a linked list containing a sequence of integers separated by 0s. The beginning and end of the list will also have `val == 0`. Merge all nodes between zeros into a single node whose value is their sum.

Example: Input: `head = [0,3,1,0,4,5,2,0]` | Output: `[4,11]` (since 3+1=4 and 4+5+2=11)

🔗 Open Problem on LeetCode (2181)